Ex:no:2

Study of different types of Network cables

Aim:

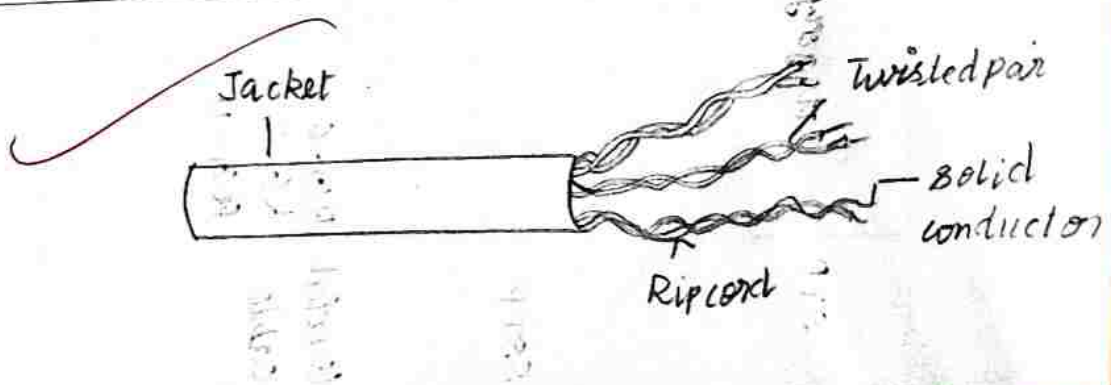
To study the different types of Network Cables.

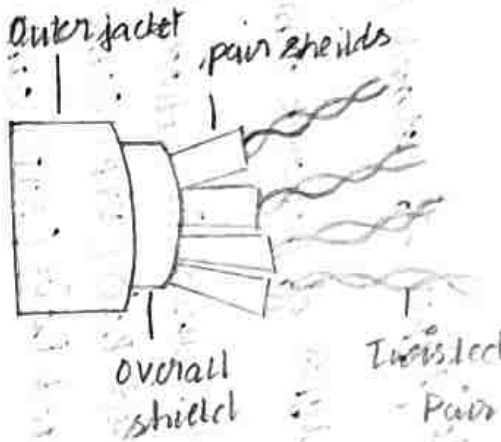
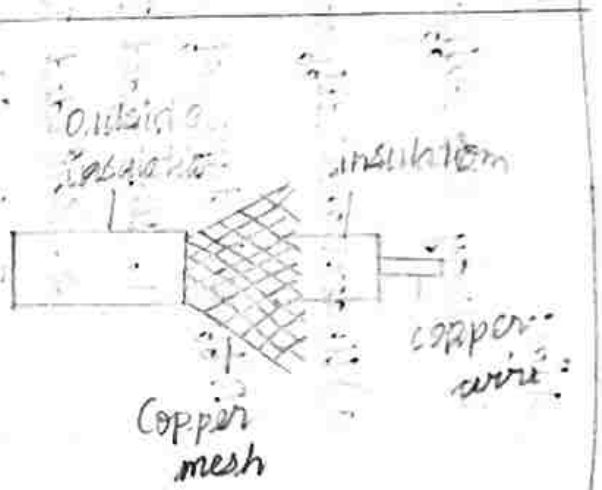
a) Understand different types of network cables.

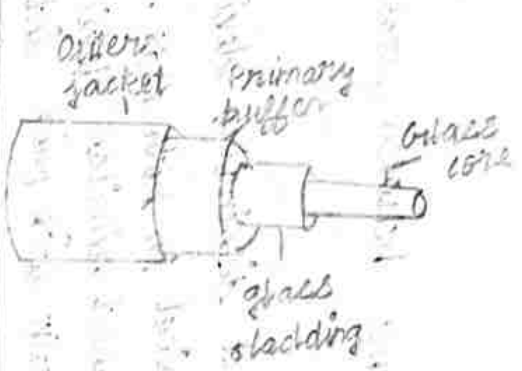
Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre Optic Cable.

Cable	Category	Max Data trans.	Advn / disadvn	Use
UTP	category 3	10 bps	Advn * cheaper in cost * Easy to install as they have a smaller overall diameter Disadvn * More prone to EMI and noise	10Base-T Ethernet
	category 5	Upto 100 Mbps		Fast Ethernet Gigabit Ethernet
	Category 5e	1Gbps		Fast Ethernet, Gigabit Ethernet



STP SSTP	Category 6, 6a	10 Gbps 10 Gbps	Advn * Shielded * Faster than UTP * Less susceptible to noise & interference disadvn * Expensive * Greater installation effort	Gigabit Ethernet 10G Ethernet (55m) Widely used in data centres Gigabit Ethernet 10G Ethernet (100m)	
Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	* High bandwidth * Immune to interference * Low loss bandwidth * Versatile disadvn * Limited distance * Cost * Size is bulky	Speed of signal is 500 m Television network high speed internet connections.	

Fibre optics cable	single mode Multimode	100 Mbps	Advn * High speed * High bandwidth * High security * Long distance Disadvn * Expensive * Required skilled Installers	* Maximum distance of fibre optics cable is around 100 meters.	

Ex.

27/

Student Observation

1. What is the difference between cross cable and straight cable?

Straight cables connect dissimilar devices by keeping the pin order identical on both ends.
Crossover cables connect similar devices by swapping the transmit and receive wire pairs internally.

2. Which type of cable is used to connect two PC?

A crossover cable is used to connect two PCs directly to each other.

3. Which type cable is used to connect a router/switch to your PC?

A straight cable is used to connect a router or a switch to your PC.

4. Find out the category of twisted pair cable is used in your Rit lab to connect the PC to the network socket.

CAT6

5.

Result:

Thus different types of cable networks are learned successfully